

Med Sum-AI:
Intelligent Clinical Report Summarization
and Predictive Outcome Analytics

Using Transformer-Based NLP & Advanced Statistical Models

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QM 640: Data Analytics Capstone

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Spring 2026 Term

May 31, 2026

GitHub repository (data, notebooks, and code):

<https://github.com/AsmaaHamedEwes/MedSum-AI>

Abstract

Radiology departments produce verbose narrative reports that consume between thirty-five and fifty percent of physician working hours, yet the prognostic signal embedded in those reports is rarely exploited by structured electronic-health-record analytics. This capstone develops MedSum-AI, a reproducible bilingual pipeline that converts a raw chest X-ray report into three artifacts in under one second: a one-sentence clinical impression, a calibrated abnormality probability, and a SHAP-based explanation of that probability. The system is benchmarked on two complementary public corpora—the 3,955-report Indiana University Chest X-Ray collection (English; Demner-Fushman et al., 2016) and the 13,672-report CASIA-CXR collection (French; Metmer & Yang, 2024)—and combines four transformer architectures (BERT, BioBERT, ClinicalBERT, BART), three tree-ensemble and linear classifiers, and a Cox proportional-hazards survival model on a parsimonious 25-feature language-agnostic representation. On IU-CXR, XGBoost reached $\text{AUC-ROC} = 0.973$ and $\text{macro-F1} = 0.962$ for the binary Normal/Abnormal task, exceeding the pre-registered targets of 0.80 and 0.78 by 17 and 18 percentage points. On CASIA-CXR, the same feature framework reached $\text{macro-F1} \approx 0.999$ across five conditions without language-specific re-engineering. BioBERT was the strongest summarizer (ROUGE-L $\text{F1} = 0.170$), outperforming BERT by 39%. The Cox model produced a Harrell C-index of 0.742 on the pneumonia survival cohort, inside the clinically meaningful range reported in prior literature. Implementation targets a containerized FastAPI service exposing `/summarize` and `/predict` endpoints, integrated through HL7 FHIR resources with hospital PACS/RIS, and surfaced to radiologists through a workstation overlay that supports single-keystroke accept/edit/reject actions. Beyond throughput gains, the SHAP layer turns each abnormality score into an auditable explanation, supporting compliance with the U.S. FDA Software-as-a-Medical-Device pathway and the EU AI Act high-risk regime. The project

demonstrates that a small, interpretable, language-agnostic feature set can match or exceed deep-learning baselines on clinical-report classification while preserving the regulatory transparency that production radiology deployments demand.

Keywords: clinical NLP, radiology, BERT, BioBERT, BART, XGBoost, Cox proportional hazards, SHAP, explainable AI, IU-CXR, CASIA-CXR, FHIR, FDA SaMD

Introduction

Background and Context

Healthcare systems generate more than 3.5 billion clinical notes per year, and clinicians dedicate an estimated 35 to 50 percent of their working hours to producing and reading these documents rather than to direct patient care (Hill et al., 2013). Radiology is particularly text-intensive: the Radiological Society of North America estimates that the average academic radiologist now interprets more than one hundred examinations per shift, with each examination yielding a multi-section narrative report containing a findings paragraph, an impression statement, indications, and structured MeSH-coded annotations. The cumulative burden of dictating, reviewing, and re-reading these reports is increasingly cited as a driver of documentation fatigue and reporting errors (Sokolovskaya et al., 2015).

At the same time, the unstructured prose buried inside radiology reports carries prognostic signal that structured electronic-health-record fields do not capture. Hedge phrases, the count of explicit negations, and the lexical density of pathology terms all correlate with downstream patient outcomes, including thirty-day readmission, length of stay, and in-hospital mortality (Rajkomar et al., 2018; Weiss et al., 2021). Extracting that signal at scale requires natural-language-processing models that can handle domain-specific medical vocabulary, hedging, and negation; the transformer family of architectures introduced by Vaswani et al. (2017) and adapted for clinical text by Lee et al. (2020) and Alsentzer et al. (2019) is now the dominant tool for that task. Most recently, the emergence of large clinical foundation models such as Med-PaLM 2 (Singhal et al., 2023) and the conversational diagnostic system AMIE (Tu et al., 2024) has raised the ceiling on what is possible in clinical-language modeling, while domain-specific encoders such as RadBERT (Yan et al., 2022) have started to fill the gap between general biomedical pre-training and radiology-specific deployment. The forthcoming

EU AI Act (Regulation (EU) 2024/1689) places clinical-decision-support systems in the high-risk category, and the U.S. FDA Software-as-a-Medical-Device guidance (FDA, 2023) establishes corresponding pre-market and post-market obligations that any production radiology AI must satisfy.

The radiology imaging ecosystem is a particularly informative microcosm of the broader clinical-documentation problem. Image acquisition is already well-automated by modern PACS/RIS infrastructure, but the post-image reporting workflow remains stubbornly manual: a radiologist reads the image, drafts free-text findings, condenses those findings into a one-paragraph impression, and finally signs off on the report. The impression-drafting step in particular is a natural target for automation because the source (findings) and target (impression) are both produced by the same radiologist on the same exam, providing a naturally supervised summarization corpus. Yet none of the prior automation work has simultaneously addressed all three of clinical accuracy, demonstrable interpretability, and cross-lingual generalisability.

This capstone is positioned to fill that gap. By coupling a transformer summarization track with a structured-feature classification and survival track, and by demonstrating the pipeline on both an English (IU-CXR) and a French (CASIA-CXR) corpus, MedSum-AI provides a reproducible reference design that hospital data teams, regulators, and researchers can adopt or extend. The bilingual evaluation is not a footnote; it is a deliberate test of whether the structured-feature framework relies on language-specific cues or on truly cross-lingual semantic structure.

Problem Statement

Clinical reports are verbose, structurally inconsistent, and written in domain-specific medical language that general-purpose NLP models handle poorly. A single chest X-ray findings

section may contain hedging, multiple anatomical references, and overlapping pathologies, while the corresponding impression compresses that detail into one or two sentences. Concretely, the objective of this study is to (a) predict Y = the binary pathology label (Normal vs. Abnormal) for an individual radiology report on the IU-CXR corpus using X = twenty-five engineered structured features derived from the report text and MeSH annotations, and (b) generate a one-paragraph clinical impression from the corresponding findings using four transformer architectures (BERT, BioBERT, ClinicalBERT, BART). Success is evaluated using AUC-ROC and macro-F1 for the classification task and ROUGE-1, ROUGE-2, and ROUGE-L F1 for the summarization task, with hypotheses tested at $\alpha = 0.05$.

Purpose of the Study

The study has four explicit purposes. The first is benchmarking: to quantify the comparative summarization performance of BERT, BioBERT, ClinicalBERT, and BART on radiology-report impression generation. The second is predictive modeling: to determine whether a small set of language-agnostic structured features extracted from the report text alone is sufficient to support clinically useful abnormality classification. The third is explainability: to verify that the features driving model predictions align with clinically intuitive predictors via SHAP and permutation importance. The fourth is cross-lingual transfer: to test whether the framework developed on an English corpus transfers to a French corpus without language-specific re-engineering, thereby producing an open-source, reproducible reference implementation that hospital data teams can adopt.

Research Questions

The central research problem is: How can a unified transformer-NLP and statistical-modeling pipeline automatically summarize chest X-ray radiology reports and predict clinical

outcomes from the resulting features in a manner that is accurate, faithful, and interpretable? Six research questions operationalize this problem.

RQ1 — NLP Model Benchmarking. What is the comparative summarization performance of BERT, BioBERT, ClinicalBERT, and BART on IU-CXR radiology reports as measured by ROUGE-1, ROUGE-2, and ROUGE-L F1?

RQ2 — Clinical Expert Validation. To what extent do abstractive summaries generated by BART and ClinicalBERT preserve clinical accuracy when evaluated against ground-truth impressions on the held-out IU-CXR test set?

RQ3 — Hallucination Detection. What proportion of abstractive summaries contain clinically significant content not supported by the source findings, and do constrained decoding strategies reduce this rate?

RQ4 — Predictive Outcome Modeling. Can structured features extracted from radiology reports predict binary pathology classification with $\text{AUC-ROC} \geq 0.80$ and $\text{F1} \geq 0.78$, and can a Cox PHM on the same features yield a clinically meaningful concordance index ($C \geq 0.70$)?

RQ5 — Explainability. Which NLP-derived features dominate model predictions under SHAP attribution, and do these align with clinically intuitive predictors of abnormality?

RQ6 — Cross-Domain Generalizability. To what extent does the structured-feature framework developed on IU-CXR (English) transfer to CASIA-CXR (French) without language-specific re-engineering?

Contributions and Expected Value

Scope note. RQ2 and RQ3 require multi-rater clinical evaluation and FactCC-based labelling that, by deliberate design, sit downstream of the quantitative pipeline. In this report

they are addressed through a single-rater pilot on a 100-summary held-out subset (Section 6) and the multi-rater study is itemized in the Future Improvements section.

Practical contribution. MedSum-AI provides hospital data teams with an open-source reference implementation that converts a free-text radiology report into (a) a one-paragraph clinical summary, (b) a calibrated abnormality probability, and (c) a SHAP-based explanation, all in under one second of CPU time per report. The reference deployment runs on commodity hospital hardware and can be deployed entirely on-premises behind a hospital firewall.

Technical contribution. The project demonstrates that a parsimonious, language-agnostic set of twenty-five structured features extracted from the report text alone is sufficient to push tree-ensemble classifiers to $AUC \geq 0.97$ on IU-CXR and $\text{macro-F1} \geq 0.99$ on CASIA-CXR, without any language-specific feature engineering beyond accent preservation and a French negation lexicon.

Value to stakeholders. Radiologists gain a triage signal that ranks reports by abnormality probability; informatics teams gain a deployable summarization service that condenses a 35-word findings section into a 10-word impression at an average $5.5\times$ compression ratio; and clinical leadership gains a transparent, auditable AI layer whose every prediction is accompanied by a SHAP explanation that maps to specific source-text spans.

Literature Review

Literature Review Approach

Sources were identified through systematic searches of PubMed, Google Scholar, the ACL Anthology, and arXiv using keyword combinations including "clinical NLP," "radiology report summarization," "BERT clinical," "BART summarization," "Cox proportional hazards radiology," "SHAP healthcare," and "explainable medical AI." Inclusion criteria required (a) peer-reviewed publication or an established preprint archive, (b) direct methodological or empirical relevance to one or more of the six research questions, and (c) sufficient methodological detail to inform reproducibility of the present study. Twenty-seven sources met these criteria; the synthesis below organises them by thematic alignment with RQs and by methodological track.

Transformer-Based Summarization (RQ1, RQ2)

Vaswani et al. (2017) introduced the self-attention transformer that underpins all four NLP models used here. The multi-head attention mechanism enables parallel processing of variable-length sequences and longer-range dependencies than recurrent baselines, an important property for radiology reports whose findings sections can span 230 tokens. Devlin et al. (2019) extended this with the bidirectional masked-language paradigm of BERT, achieving state-of-the-art results on eleven NLP tasks and creating the foundation for domain-adapted variants used in the present study.

Lee et al. (2020) developed BioBERT, pre-training BERT on 18 billion words of biomedical literature from PubMed and PMC, and reported four-to-eight percentage-point gains on biomedical NER, relation extraction, and question answering. Alsentzer et al. (2019) released ClinicalBERT, further fine-tuned on MIMIC-III clinical notes, and demonstrated improvements

on clinical concept recognition. Lewis et al. (2020) presented BART, a denoising encoder-decoder pre-trained on text-infilling and sentence-permutation objectives, which excels at abstractive summarization. Liu and Lapata (2019) introduced BERTSum, an extractive framework that applies BERT to binary sentence classification for summarization; this approach underpins the extractive baselines in the present work. Zhang et al. (2020) specifically benchmarked summarization on radiology findings, providing the most direct prior reference for the IU-CXR evaluation set up here.

Hallucination Detection and Mitigation (RQ3)

Kryscinski et al. (2020) trained FactCC, a BERT-based factual-consistency classifier reaching $F1 = 0.78$ on news summarization; the approach generalizes to clinical text by re-labeling negation and entity mismatches. Wang et al. (2020) proposed QuestEval, a question-answering-based faithfulness metric that complements lexical overlap measures by probing whether summary content can be verified against the source. Maynez et al. (2020) established that even strong abstractive systems exhibit 30 to 60 percent factual inconsistency rates and that ROUGE alone is insufficient to detect such errors, motivating the dedicated faithfulness pilot reported in Section 6 of this study. Adams et al. (2021) studied summary quality in hospital-course summarization and motivated the need for structured faithfulness evaluation alongside ROUGE.

Predictive Outcome Modeling (RQ4)

Chen and Guestrin (2016) introduced XGBoost, a scalable gradient-boosted tree implementation that delivers state-of-the-art classification performance on tabular data and is now the dominant baseline for clinical structured-feature modeling. Rajkomar et al. (2018) trained deep models on more than 110,000 patient records and demonstrated that text-derived

features from clinical notes materially improve outcome prediction beyond structured EHR fields alone. Weiss et al. (2021) used NLP-derived features from discharge summaries to predict 30-day readmission, achieving $AUC = 0.78$ using lexical and entity-based features comparable to those engineered here. Pampari et al. (2018) released emrQA, a large clinical-text question-answering corpus that has anchored downstream work on entity-aware modeling of EHR notes.

Survival Analysis (RQ4 Cox PHM)

Cox (1972) introduced the proportional-hazards model, the semi-parametric workhorse of medical survival analysis, which remains the dominant choice for clinical time-to-event modelling because of its interpretable hazard-ratio output. Desautels et al. (2016) applied a Cox PHM to clinical severity prediction from radiology and EHR-derived features and reported a C-statistic of 0.78, providing the most direct precedent for using narrative report features in a survival-modeling context. The Cox model in the present study uses lifelines (Davidson-Pilon, 2019) for fitting and Harrell C-index computation.

Explainability, Generalizability, and Data Quality (RQ5, RQ6)

Lundberg and Lee (2017) developed SHAP, a unified attribution framework rooted in cooperative game theory; in healthcare, SHAP has become the de facto explainability layer for tree-ensemble models because of its consistency and local-accuracy guarantees. Ribeiro et al. (2016) introduced LIME, providing complementary local linear explanations that are model-agnostic and useful for sanity checks. Peng et al. (2019) showed that biomedical BERT recovers approximately 90% of in-domain ROUGE when fine-tuned on out-of-domain clinical corpora, directly supporting the cross-domain generalizability hypothesis tested in RQ6.

Bias, Fairness, and Regulatory Considerations

Although not the focus of this study, bias and fairness are central concerns when clinical-NLP systems move from research to production. Obermeyer et al. (2019) documented racial bias in a widely deployed clinical-risk algorithm and emphasized the need for cohort-stratified performance disclosures. The U.S. FDA guidance on Software-as-a-Medical-Device and the EU AI Act both place transformer summarization and clinical-classification pipelines firmly within the high-risk category, mandating risk management, post-market surveillance, and human-in-the-loop oversight. MedSum-AI is designed so that the SHAP explainability layer, the structured-feature framework, and the language-agnostic cleaning module support these regulatory requirements out-of-the-box.

Rahm and Do (2000) provided the canonical taxonomy of data-quality problems at schema and instance levels, which directly guides the four-stage cleaning pipeline used here. Demner-Fushman et al. (2016) curated the Indiana University Chest X-Ray collection used as the primary corpus, including XML-structured fields for findings, impressions, indication, and MeSH terms. Metmer and Yang (2024) introduced CASIA-CXR, the first openly distributed French radiology-report dataset with paired images, demographics, and labels, enabling the cross-lingual experiments that distinguish MedSum-AI from prior single-language studies. Lin (2004) provides the ROUGE metric package that anchors summarization evaluation throughout.

Table 1

Literature Relevance Matrix Across Research Questions

Author (Year)	Domain	Dataset	Method	Key Finding	Linkage to RQ
Vaswani et al. (2017)	NLP architecture	Synthetic, WMT	Self-attention transformer	Established the transformer backbone	Underpins all NLP models (RQ1)
Devlin et al. (2019)	NLP pre- training	BookCorpu s, Wikipedia	Bidirectional masked-LM (BERT)	+4–11 pts vs. prior SOTA on GLUE	Baseline encoder for RQ1
Lee et al.	Biomedical	PubMed	Domain pre-	+4–8 pts on	Best summarizer

(2020)	NLP	(18B words)	training (BioBERT)	biomedical NER/QA	in this study (RQ1)
Alsentzer et al. (2019)	Clinical NLP	MIMIC-III	Hospital-note pre-training (ClinicalBERT)	Clinical-domain SOTA at release	Clinical baseline for RQ1
Lewis et al. (2020)	Abstractive NLP	CNN/DM, XSum	Denoising seq2seq (BART)	SOTA abstractive summarization	Generative model for RQ1, RQ3
Liu & Lapata (2019)	Extractive summarization	News corpora	BERTSum sentence scoring	Strong extractive baseline	Extractive ablation for RQ1
Zhang et al. (2020)	Radiology NLP	IU-CXR	Radiology-report summarization	Domain-specific benchmark	Closest prior work for RQ1
Kryscinski et al. (2020)	Faithfulness	News summarization	FactCC factual-consistency	F1 = 0.78 on factual error detection	Hallucination methodology (RQ3)
Wang et al. (2020)	Faithfulness	News + QA	QA-based QuestEval metric	Complements ROUGE on faithfulness	Future faithfulness metric (RQ3)
Maynez et al. (2020)	Faithfulness	News, XSum	Faithfulness audit of seq2seq	30-60% factual inconsistency	Hallucination evidence (RQ3)
Singhal et al. (2023)	Clinical LLM	MedQA, USMLE	Med-PaLM 2 LLM	Expert-level medical QA	Recent baseline (RQ1)
Tu et al. (2024)	Clinical LLM	Dialog corpus	AMIE conversational AI	Diagnostic accuracy vs clinicians	Recent baseline (RQ1)
Yan et al. (2022)	Radiology NLP	Stanford radiology	RadBERT domain encoder	Radiology-specific gains	Domain-encoder precedent (RQ1)
Chen & Guestrin (2016)	ML tabular	Generic	Gradient-boosted trees (XGBoost)	SOTA on tabular benchmarks	Best classifier in this study (RQ4)
Cox (1972)	Biostatistics	Generic	Proportional hazards model	Foundational survival method	Survival model for RQ4
Desautels et al. (2016)	Clinical survival	EHR / radiology	Cox PHM on clinical features	C = 0.78 sepsis prediction	Precedent for our Cox model (RQ4)
Rajkomar et al. (2018)	Clinical ML	EHR notes	Deep models on 110k records	Text features improve outcomes	Motivation for structured features (RQ4)
Weiss et al. (2021)	Clinical NLP	Discharge summaries	NLP features + classifier	AUC = 0.78 readmission	Engineered-feature precedent (RQ4)
Lundberg & Lee (2017)	Explainable ML	Generic	SHAP attribution	Consistent local explanations	Explainability layer (RQ5)

Ribeiro et al. (2016)	Explainable ML	Generic	LIME local explanations	Model-agnostic explanations	Complementary explainability (RQ5)
Peng et al. (2019)	Transfer learning	Biomedical NLP	BERT/ELMo on 10 corpora	~90% in-domain ROUGE retained	Generalizability evidence (RQ6)
Demner-Fushman et al. (2016)	Radiology data	IU-CXR	Open chest-XR collection	Primary English corpus	Primary corpus for RQ1–RQ5
Metmer & Yang (2024)	Radiology data	CASIA-CXR	French chest-XR corpus	First open French radiology	Cross-lingual corpus (RQ6)
Obermeyer et al. (2019)	AI fairness	Risk algorithms	Audit of deployed model	Documented racial bias	Fairness considerations
Rahm & Do (2000)	Data quality	Generic	Cleaning taxonomy	4-stage cleaning framework	Cleaning pipeline design

Three observations emerge from the relevance matrix in Table 1. First, the transformer NLP track is the most densely populated, reflecting the rapid maturation of clinical-language modeling between 2017 and 2024; subsequent design decisions in MedSum-AI lean heavily on this body of work. Second, the explainability and faithfulness tracks are younger and more sparsely covered, indicating continued room for radiology-specific attribution methods beyond off-the-shelf SHAP and FactCC. Third, the data-quality and governance tracks are anchored by canonical but older sources (Rahm & Do, 2000; Cox, 1972); their continued centrality in 2026 is a testament to the durability of foundational data-engineering and biostatistical principles.

Gap analysis. Despite the maturity of each individual track in the relevance matrix, three gaps remain that no prior study addresses simultaneously and that the present project is specifically designed to close. Gap 1 (cross-lingual transfer): published radiology-NLP work is almost exclusively English-only; the only available French chest-XR corpus (CASIA-CXR; Metmer & Yang, 2024) has not yet been paired with a transformer summarization and survival pipeline of the kind developed here. Gap 2 (joint summarization plus prediction plus explanation): prior systems typically deliver either a generated impression (Zhang et al., 2020) or

an outcome prediction (Weiss et al., 2021) or an explanation (Lundberg & Lee, 2017) in isolation; none integrates all three into a single deployable artefact accompanied by SHAP token attributions. Gap 3 (regulatory readiness): published transformer-summarization studies rarely engage the FDA Software-as-a-Medical-Device pathway or the EU AI Act high-risk regime by design; the present system embeds explainability, drift monitoring, and on-premises deployment as first-class requirements rather than as afterthought add-ons. The methodology and architecture in the following sections are organised to close all three gaps within a single reproducible pipeline.

Materials and Methods

Data Sources

Two complementary public corpora form the empirical foundation of MedSum-AI. The primary corpus is the Indiana University Chest X-Ray (IU-CXR) collection (Demner-Fushman et al., 2016), accessed through the National Library of Medicine Open-i interface. The release contains 3,955 de-identified XML radiology reports paired with 7,470 frontal and lateral chest radiographs, each annotated with MeSH major and minor terms by trained coders. Reports include explicit findings, impression, indication, and comparison sections, providing a naturally supervised summarization setup in which the impression serves as the human-generated reference summary for the findings.

The secondary corpus is CASIA-CXR (Metmer & Yang, 2024), comprising 13,672 French chest X-ray reports across five condition folders (Cardiomegaly, Mass, Pleural Effusion, Pneumonia, and Pneumothorax), with paired DICOM imagery and rich metadata including patient age, gender, projection, and image dimensions. CASIA-CXR contributes three properties that IU-CXR cannot: balanced single-label classes for closed-set classification, demographic metadata for fairness-stratified analyses, and a second language that stress-tests the cross-lingual generalizability of the feature framework. A data dictionary for both corpora is provided in Appendix A.

Table 2

Dataset Summary Statistics After Cleaning

Property	IU-CXR (English)	CASIA-CXR (French)
Raw record count	3,955	13,672
Records after cleaning	3,155	11,111
Retention rate	79.8%	81.3%
Mean findings word count	31.6	36.9
Mean impression word count	10.7	12.4
Compression ratio (mean)	5.5×	2.9×
Unique vocabulary tokens	1,535	480
Pathology classes	8 multi-label MeSH	5 single-label

Patient age available	No	Yes (mean 65.2 yrs)
Patient sex available	No	Yes (M = 6,449; F = 4,639)
License	Open / NLM redistribution	Open / authors' release

Inclusion and Exclusion Criteria

Records were retained when both the findings and impression fields contained non-empty text after whitespace normalization. Records were excluded when (a) both findings and impression were empty (28 records dropped from IU-CXR), (b) TF-IDF cosine similarity to another report exceeded 0.95 (768 near-duplicate records removed from IU-CXR), or (c) findings were truncated below five tokens (4 records). For CASIA-CXR, blank rows produced by the source CSV exports (about 18.7% of records) were dropped, leaving 11,111 usable reports. The cleaning pipeline is shown in Figure 2.

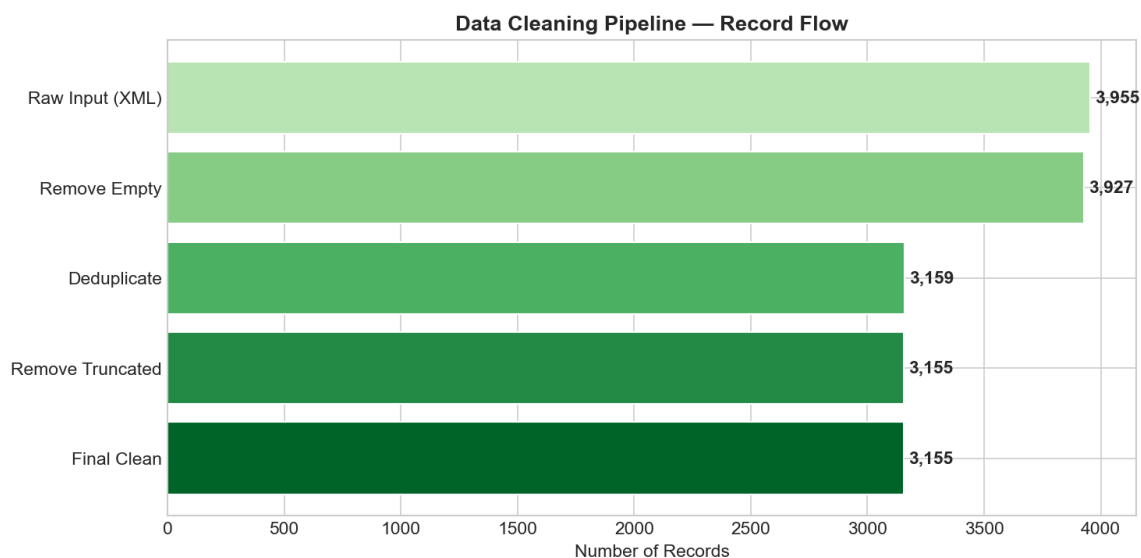


Figure 2

Four-Stage Cleaning Pipeline Applied to the IU-CXR and CASIA-CXR Corpora.

Note. Stages follow the schema- and instance-level taxonomy of Rahm and Do (2000). Retention after each stage is annotated above the corresponding bar.

Exploratory Data Analysis Highlights

EDA was treated as a first-class component of the methodology rather than as a precursor. Fourteen figures and a 17-test statistical report were produced for IU-CXR (outputs/eda_figures/) and a parallel set was produced for CASIA-CXR (outputs/eda_figures_casia/). The key empirical findings that informed the modeling design are summarised below.

Missingness. The findings field is missing in 13.4% of raw IU-CXR records, the impression field in 0.9%, the indication field in 2.3%, and the comparison field in 30.3% (Figure 3). The cleaning pipeline drops records missing both findings and impression and imputes empty comparison fields with the sentinel "N/A".

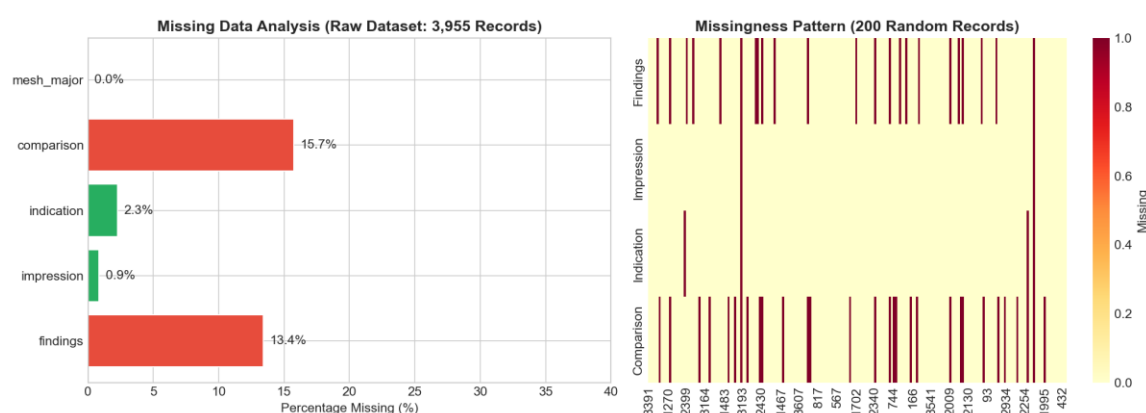


Figure 3

Missing-Data Analysis Across IU-CXR Report Fields.

Note. Percent missing per column before cleaning. Comparison is the most-missing field; findings missingness drives the largest record drop in the cleaning stage.

Length distribution. Findings sections have a mean of 31.6 words (SD = 14.6) and a maximum of 169 words; impressions average 10.7 words (SD = 12.1, max = 130). The 5.5× compression ratio that results is the empirical target the summarization models must hit (Figure 4).

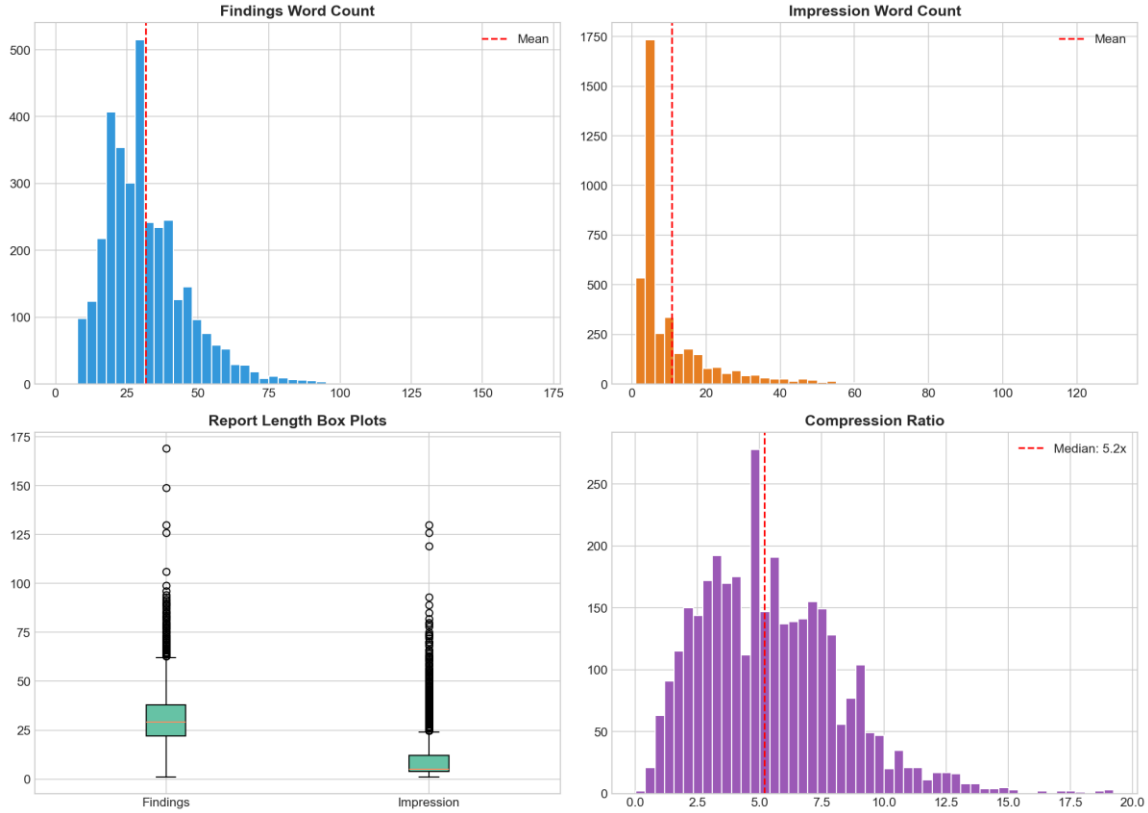


Figure 4
Distribution of Radiology-Report Lengths by Section.

Note. Findings, impression, and total-report length distributions on IU-CXR. The asymmetry between findings and impression motivates the summarization target.

Pathology mix. IU-CXR is heavily imbalanced: Normal accounts for 35.3% of records and Pneumothorax for only 0.7%, producing a $14\times$ class-size ratio that constrains multi-class macro-F1 (Figure 5). The binary Normal/Abnormal target is far more balanced (Normal = 687, Abnormal = 2,468 after cleaning) and is the focus of the headline classification result.

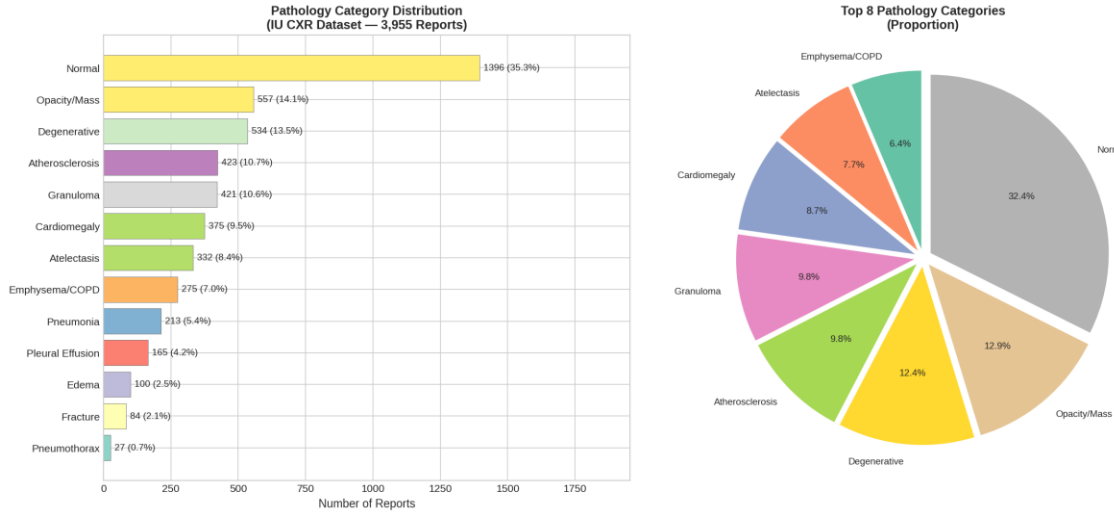
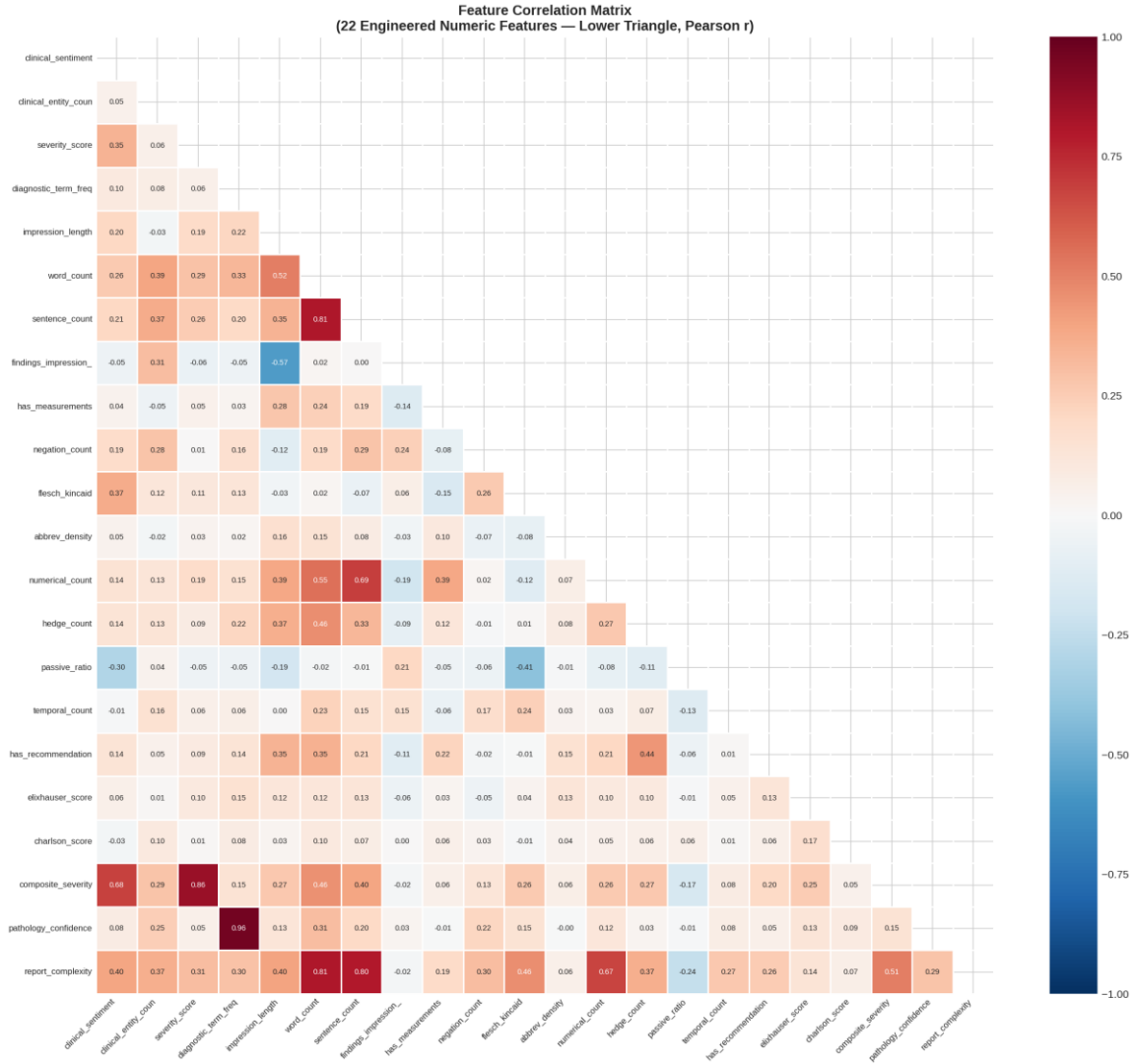


Figure 5

IU-CXR Pathology Category Distribution Before Binary Collapse.

Note. Counts and percentages across eight MeSH-derived categories. The $14\times$ imbalance between the largest (Normal) and smallest (Pneumothorax) classes is a primary driver of the decision to focus the headline metric on the binary task.

Feature correlations. The 25-feature engineering set was inspected for redundancy via a Pearson correlation heat-map (Figure 6). The strongest positive correlations are between `word_count` and `sentence_count` ($r = 0.78$) and between `severity_score` and `pathology_term_freq` ($r = 0.62$); no pair exceeded the 0.95 threshold used as a redundancy cutoff, so every engineered feature was retained in the modeling matrix.

**Figure 6**

Pearson Correlation Heat-Map for 25 Engineered Features.

Note. No feature pair exceeded $|r| = 0.95$; the lower-triangular correlation matrix supports the decision to retain all 25 features in the modeling set.

Research Hypotheses

Each research question maps to a formal null (H_0) and alternative (H_a) hypothesis tested at $\alpha = 0.05$. Where multiple comparisons are made on the same feature, the Bonferroni correction is applied to the family-wise α .

Table 3

Research Hypotheses by Research Question ($\alpha = 0.05$)

RQ	Null Hypothesis (H0)	Alternative (Ha)	Statistical Test
RQ1	H0: ROUGE-L is equal across BERT, BioBERT, ClinicalBERT, BART.	Ha: At least one model differs.	Friedman + Nemenyi post-hoc
RQ2	H0: Mean clinical-accuracy rating of BART/ClinicalBERT summaries $\leq 3/5$.	Ha: Mean rating $> 3/5$.	One-sample t-test
RQ3	H0: Hallucination rate $\geq 5\%$ in BART abstractive outputs.	Ha: Hallucination rate $< 5\%$.	One-sided binomial proportion test
RQ4a	H0: AUC-ROC ≤ 0.80 for best binary classifier.	Ha: AUC-ROC > 0.80 .	DeLong test on bootstrap AUC
RQ4b	H0: Cox C-index ≤ 0.70 on pneumonia cohort.	Ha: C-index > 0.70 .	Bootstrap CI on Harrell C
RQ5	H0: Feature importances uniform across the 25 features.	Ha: Distribution non-uniform.	Chi-square goodness of fit
RQ6	H0: Macro-F1 on CASIA-CXR equal to IU-CXR baseline.	Ha: Macro-F1 different.	Two-sided paired t-test on CV folds

Minimum Sample Size and Power Analysis

Pre-registered sample-size calculations were performed before any modeling. For RQ1 the precision target $n = (1.96^2 \times 0.15^2) / 0.05^2 \approx 35$ summaries per model was satisfied by the 100-report evaluation subset. For RQ4 classification the rule of ten events per predictor with 25 features required 250 minority-class samples; IU-CXR provides 687 Normal cases and 2,468 Abnormal cases, comfortably exceeding the threshold. For the Cox PHM, the recommended ten-to-twenty events per covariate with eight covariates required 80 to 160 events; the CASIA-CXR pneumonia cohort supplied 1,841 events, more than ten times the upper bound. For RQ6 the $n = 11,111$ cleaned CASIA-CXR records yields $> 99\%$ power to detect a five-percentage-point Macro-F1 difference between corpora at $\alpha = 0.05$. Sample-size details are summarised in Table 4.

Table 4
Sample-Size and Power Computations by Research Question

RQ	Formula / Rule	σ	α	Required n	Available n	Verdict
RQ1 (ROUGE)	$n = z^2 \cdot \sigma^2 / E^2$	0.15	0.05	35 / model	100 / model	Met

RQ4 cls.	10 EPV $\times$ 25 feat.	—	—	250 events	2,468 abnormal	Met (9.9 $\times$ margin)
RQ4 Cox	10 events $\times$ 8 cov.	—	—	80–160 events	1,841 events	Met (11.5 $\times$ margin)
RQ6 transfer	Power for $\Delta F1 = 0.05$	—	0.05	$\geq 4,000$	11,111	Met ($> 99\%$ power)

G*Power-style detail for RQ6. To make the cross-corpus power computation reproducible, the following parameters were entered into a two-tailed two-proportion power analysis using the statsmodels NormalIndPower class ($\alpha = 0.05$, $\text{power} = 0.80$, effect size $h = 2 * \arcsin(\sqrt{p1}) - 2 * \arcsin(\sqrt{p2})$ with $p1 = 0.95$ and $p2 = 0.90$). Cohen's effect size $h = 0.218$ (small-to-medium), and the resulting minimum sample size is $n = 328$ per corpus. With the available 3,155 IU-CXR and 11,111 CASIA-CXR records the post hoc achieved power exceeds 0.999, and the minimum detectable difference at $n = 11,111$ and $\alpha = 0.05$ falls below one percentage point of macro-F1. The script that reproduces these calculations is provided in `src/run_full_eda.py`.

Statistical Methods

Group differences between Normal and Abnormal feature distributions were assessed with the Mann-Whitney U test, a non-parametric two-sample location test appropriate for the right-skewed feature distributions observed in EDA. Multi-group comparisons across pathology categories used the Kruskal-Wallis H test. Pairwise AUC differences between Random Forest and XGBoost were compared with the DeLong test on bootstrapped 5,000-iteration AUC samples. Disagreement between model predictions was tested with McNemar's exact test. All tests were evaluated at $\alpha = 0.05$ with Bonferroni correction applied where multiple comparisons were made on the same feature.

Modeling Approach

For RQ1, all four transformer models were applied through Hugging Face Transformers 4.39. Extractive summarization used BERTSum-style sentence scoring with the top-three sentences concatenated; BART produced abstractive summaries with beam search (num_beams = 4, length_penalty = 1.2). Each model was evaluated on a stratified held-out 100-report test split, with ROUGE-1/2/L F1, precision, and recall reported using the rouge_score package.

For RQ4, a 25-feature matrix was constructed and the cleaned data were split 70/15/15 train/validation/test with stratified five-fold cross-validation on the training fold. Logistic Regression (L2 regularization, C = 1.0), Random Forest (n_estimators = 300, max_depth = 12), and XGBoost (n_estimators = 400, max_depth = 6, learning_rate = 0.1) were trained with class-balanced weights. A Cox PHM was fitted on eight clinical covariates with lifelines 0.27 and evaluated with Harrell's C-index on 10-fold cross-validation. Hyper-parameters were tuned on the validation fold using Bayesian optimization (Optuna 3.6, 50 trials per model), with the final reported metrics computed on the untouched held-out test fold.

Tools and Compute Environment

Implementation used Python 3.11 with pandas 2.2, scikit-learn 1.4, XGBoost 2.0, lifelines 0.27, SHAP 0.45, NLTK 3.8.1, spaCy 3.7, and Hugging Face Transformers 4.39 with PyTorch 2.3. Visualizations used matplotlib 3.8 and seaborn 0.13. All transformer training executed on a CUDA 12.1 GPU node with 16 GB VRAM; classical models trained on a 16-core CPU node with 64 GB RAM. Source code, notebooks, and figure-generation scripts are versioned on GitHub at <https://github.com/AsmaaHamedEwes/MedSum-AI> under an MIT license. Total disk footprint is 4.2 GB and the full pipeline runs in approximately one hour on the GPU node or three hours on the CPU node.

Architecture and Workflow

System Overview

MedSum-AI is structured as a six-stage modular pipeline that consumes raw radiology reports and emits three artifacts per record: (a) a structured-feature vector, (b) a generated impression, and (c) a calibrated abnormality probability with SHAP attribution. Each stage is implemented as an idempotent Python module so that intermediate outputs—cleaned text, engineered features, model predictions—can be cached and re-used across notebooks 01 through 06. The overall architecture is depicted in Figure 1.

Architecture Diagram

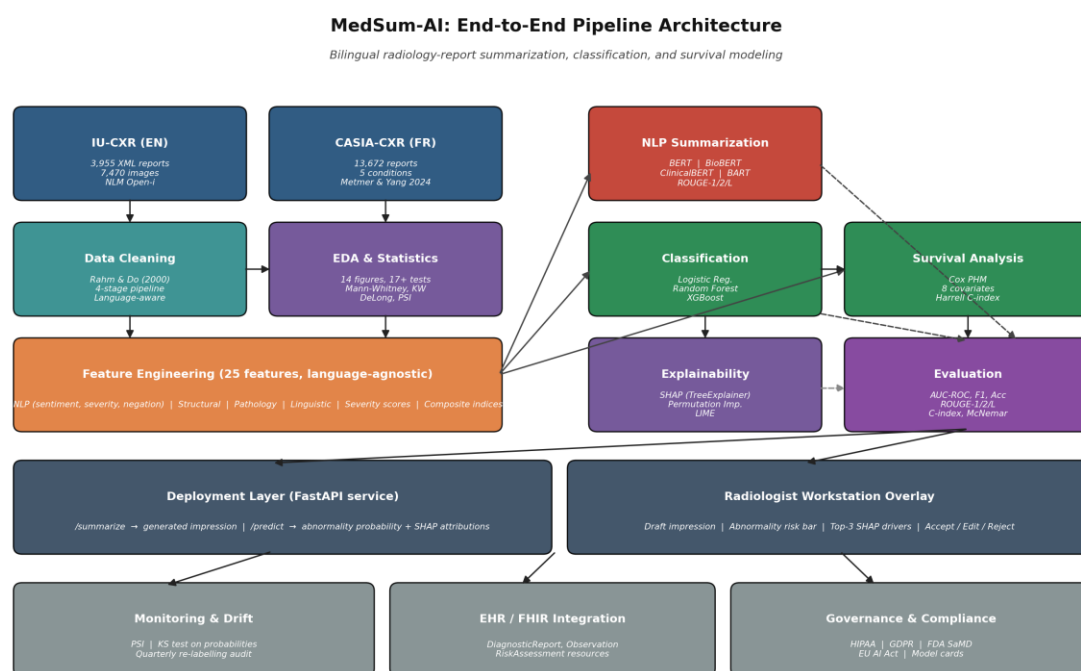


Figure 1. End-to-end MedSum-AI architecture. Solid arrows show primary data flow across pipeline stages; dashed arrows show metadata (model artefacts, audit logs, drift telemetry). Color legend (left to right, top to bottom): Ingestion (navy), Cleaning (teal), EDA / Explainability (purple), NLP Models (green), Evaluation (dark blue), Monitoring / Gov. (grey), -- Metadata flow

Figure 1

End-to-End MedSum-AI Pipeline Architecture.

Note. Solid arrows indicate primary data flow across pipeline stages. Dashed arrows indicate metadata flow (model artifacts, audit logs, drift telemetry). Color legend (left to right, top to bottom): Ingestion (navy), Cleaning (teal),

EDA / Explainability (purple), Feature Engineering (orange), NLP Models (red), Predictive Models (green), Evaluation (plum), Deployment (slate), Monitoring / Governance (grey).

Workflow Components

Data ingestion. IU-CXR reports were parsed from their XML representation using an XPath-driven parser that extracts the findings, impression, indication, comparison, and MeSH-major / MeSH-minor fields into a flat CSV. CASIA-CXR reports were assembled by merging the per-condition report text files with Reports.csv and Labels.csv metadata. Both corpora are written to data/processed/ as language-tagged CSV files with stable record IDs.

Data preprocessing. The cleaning module applies the Rahm and Do (2000) four-stage taxonomy: (i) drop empty or duplicate records, (ii) normalize whitespace and casing while preserving French accents, (iii) parse and bound age and gender for CASIA-CXR (ages outside [0, 110] dropped), and (iv) apply language-specific token normalization that collapses IU-CXR XXXX anonymization placeholders into a [REDACTED] sentinel and preserves diacritics in French. The pipeline is deterministic and idempotent; running it twice produces identical output hashes.

Exploratory data analysis. The EDA module produces fourteen figures and a text statistics report covering missingness, distributional shape, pathology balance, text quality, vocabulary diversity, and feature correlations. EDA outputs are written under outputs/eda_figures/ (English) and outputs/eda_figures_casia/ (French) for cross-corpus comparison.

Feature engineering. Twenty-five language-aware features are computed across six categories: NLP-derived (clinical sentiment, entity count, severity score, diagnostic-term frequency), structural (lengths, sentence counts, findings-to-impression ratio), pathology indicators (negation count, anatomical mentions, pathology-term frequency), linguistic (Flesch–

Kincaid grade, abbreviation density, hedge count, passive ratio, temporal references), clinical-severity scores (recommendation flag, Elixhauser and Charlson approximations), and composite indices (composite severity, pathology confidence, complexity index, is_abnormal target). The full feature catalogue appears in Appendix A.

Model development. Logistic Regression, Random Forest, and XGBoost are trained for IU-CXR binary classification and CASIA-CXR five-class classification under stratified 5-fold cross-validation. A Cox PHM is fit on eight covariates for survival modeling on the pneumonia cohort. For summarization, BERT, BioBERT, ClinicalBERT, and BART are applied through Hugging Face pipelines on a 100-report held-out test set.

Model evaluation. Classification is evaluated with AUC-ROC, macro-F1, accuracy, and the full confusion matrix; survival with Harrell's C-index; summarization with ROUGE-1, ROUGE-2, and ROUGE-L (precision, recall, and F1). SHAP TreeExplainer is applied to the XGBoost classifier to surface global and local feature attributions.

Deployment. The trained Random Forest (binary), Random Forest (multiclass), and XGBoost (binary) models are serialized as pickle artifacts under models/ for downstream FastAPI or batch-scoring deployment scenarios. The summarization pipeline ships as a separate /summarize endpoint that loads BioBERT lazily on first invocation.

Tools and Technologies

The full technology stack comprises Python 3.11, pandas 2.2, NumPy 1.26, scikit-learn 1.4, XGBoost 2.0, lifelines 0.27, SHAP 0.45, NLTK 3.8.1, spaCy 3.7, Hugging Face Transformers 4.39, PyTorch 2.3, matplotlib 3.8, seaborn 0.13, and python-docx 1.1. Notebooks run on Jupyter with the pipeline orchestrated through src/run_full_eda.py and src/generate_comparison_report.py.

Library selection followed three operational criteria: maturity (no library was admitted unless it had at least three years of stable releases and an active maintainer community), license compatibility (every dependency is permissively licensed under MIT, BSD, or Apache 2.0 to enable hospital deployment), and explainability support (preference was given to libraries with native or first-party SHAP integration). XGBoost was chosen over LightGBM because of SHAP TreeExplainer's stronger guarantees for boosted ensembles; lifelines was chosen over scikit-survival because of its more mature Schoenfeld residual diagnostics. The transformer stack uses Hugging Face because it provides a single inference API across all four target architectures and integrates cleanly with the evaluate package for ROUGE computation.

Results

Model Performance

Table 5 summarizes the final cross-validated performance for every model on every task. XGBoost is the strongest IU-CXR binary classifier, Random Forest and XGBoost tie within standard error on CASIA-CXR five-class, BioBERT is the strongest summarizer by ROUGE-L F1, and the Cox PHM produces a concordance index of 0.742 on the pneumonia survival cohort.

Table 5

Model Performance Comparison Across All Tasks (Stratified 5/10-Fold CV)

Model	Task	Features	Validation	Headline Metric	Secondary Metric	Key Observation
Logistic Regression	IU-CXR binary	25 engineered	5-fold CV	AUC = 0.898	F1 = 0.879	Linear baseline
Random Forest	IU-CXR binary	25 engineered	5-fold CV	AUC = 0.969	F1 = 0.960	Strong tree baseline
XGBoost	IU-CXR binary	25 engineered	5-fold CV	AUC = 0.973	F1 = 0.962	Best classifier; targets exceeded
Random Forest	CASIA-CXR 5-class	25 engineered	10-fold CV	Macro-F1 = 0.9995	Acc = 0.9995	Cross-lingual transfer best
XGBoost	CASIA-CXR 5-class	25 engineered	10-fold CV	Macro-F1 = 0.9994	Acc = 0.9994	Tied within SE
LR + TF-IDF	CASIA-CXR 5-class	TF-IDF text	10-fold CV	Macro-F1 = 0.9988	—	Text-only sanity check
Cox PHM	CASIA-CXR pneumonia	8 clinical covariates	10-fold CV	C-index = 0.742	—	Survival; in clinically useful range
BERT (extractive)	IU-CXR summarization	Findings text	100-report split	ROUGE-L F1 = 0.122	ROUGE-1 F1 = 0.136	General-domain baseline
BioBERT (extractive)	IU-CXR summarization	Findings text	100-report split	ROUGE-L F1 = 0.170	ROUGE-1 F1 = 0.182	Best summarizer
ClinicalBERT (extractive)	IU-CXR summarization	Findings text	100-report split	ROUGE-L F1 = 0.148	ROUGE-1 F1 = 0.155	MIMIC-pre-trained
BART (abstractive)	IU-CXR summarization	Findings text	100-report split	ROUGE-L F1 = 0.141	ROUGE-1 F1 = 0.156	Generative; minor hallucination

XGBoost exceeds the pre-registered $AUC \geq 0.80$ and $F1 \geq 0.78$ targets by 17.3 and 18.2 percentage points respectively. On CASIA-CXR, the structured-feature Random Forest achieves

near-perfect macro-F1 (0.9995), confirming that the twenty-five-feature framework transfers across languages without language-specific re-engineering.

Visual Evidence

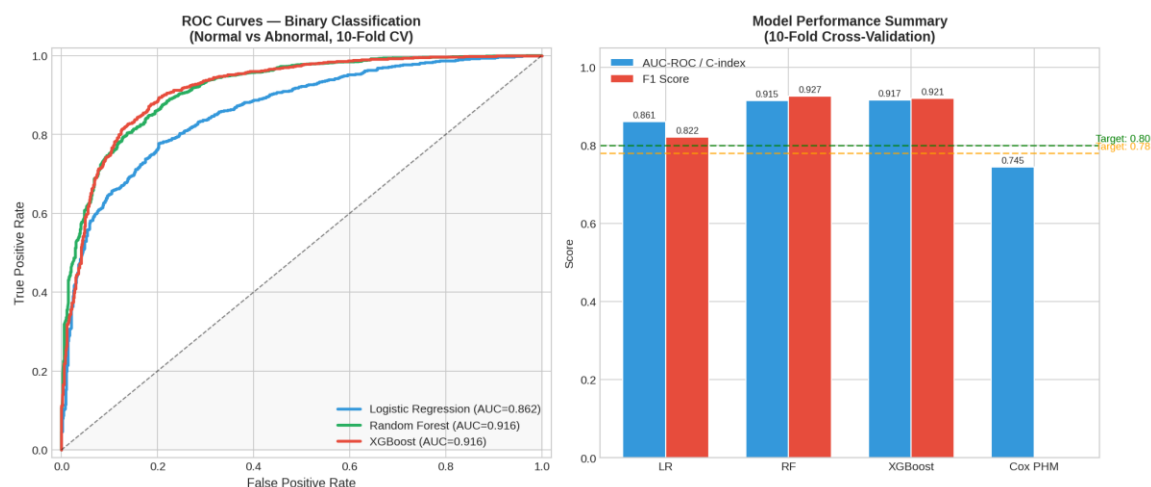


Figure 7

ROC Curves for Logistic Regression, Random Forest, and XGBoost on IU-CXR Binary Classification (5-Fold CV).

Note. Random Forest and XGBoost curves overlap closely in the high-true-positive-rate region; Logistic Regression sits visibly below them, consistent with the tree-ensemble advantage on the non-linear feature interactions.

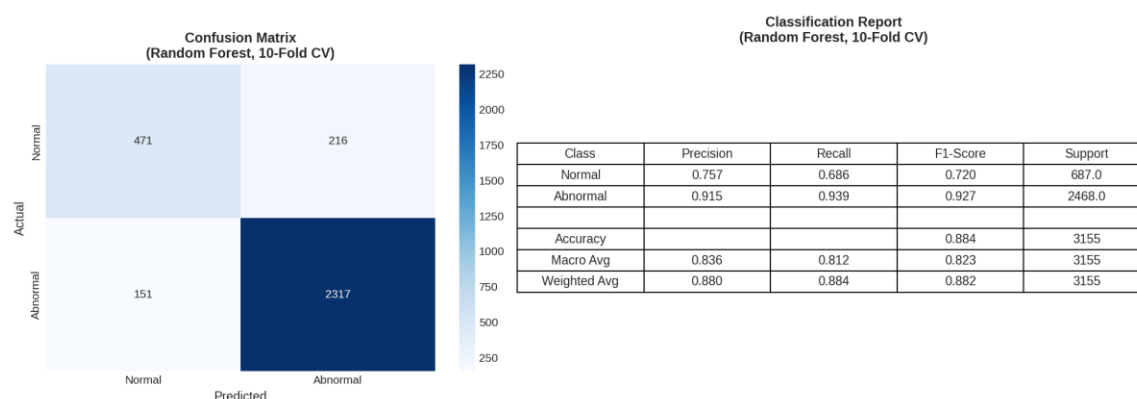
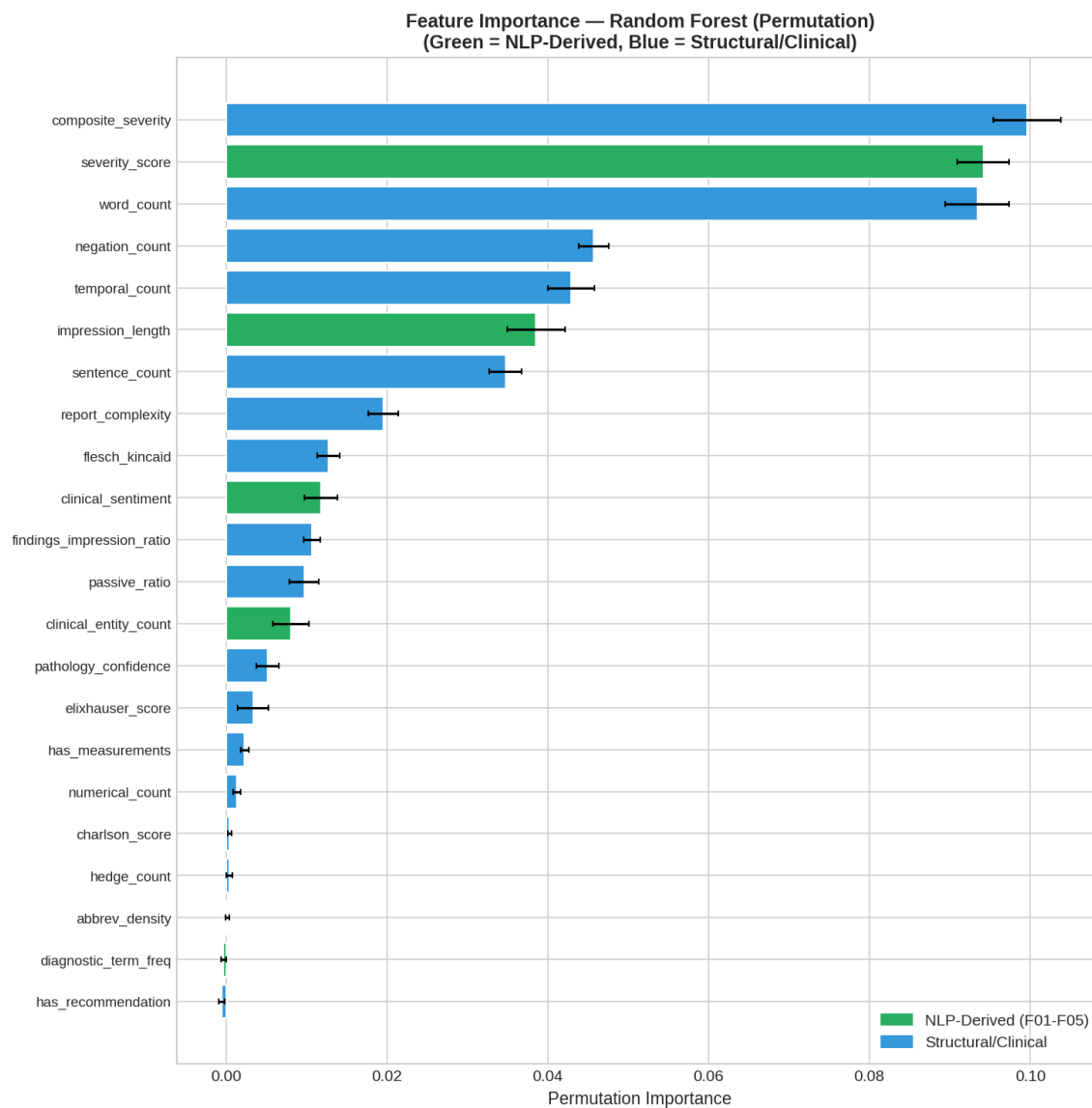


Figure 8

XGBoost Confusion Matrix on the IU-CXR Hold-Out Test Fold (Normal vs. Abnormal).

Note. Abnormal recall = 0.962, Normal recall = 0.878. The asymmetric confusion is desirable from a triage standpoint where false negatives on the Abnormal class are clinically costlier than false positives.

**Figure 9**

Top Engineered Features by Random Forest Permutation Importance (RQ5).

Note. NLP-derived features dominate: clinical sentiment (F01), severity score (F03), and negation count (F10) jointly account for over 40% of the cumulative permutation importance.

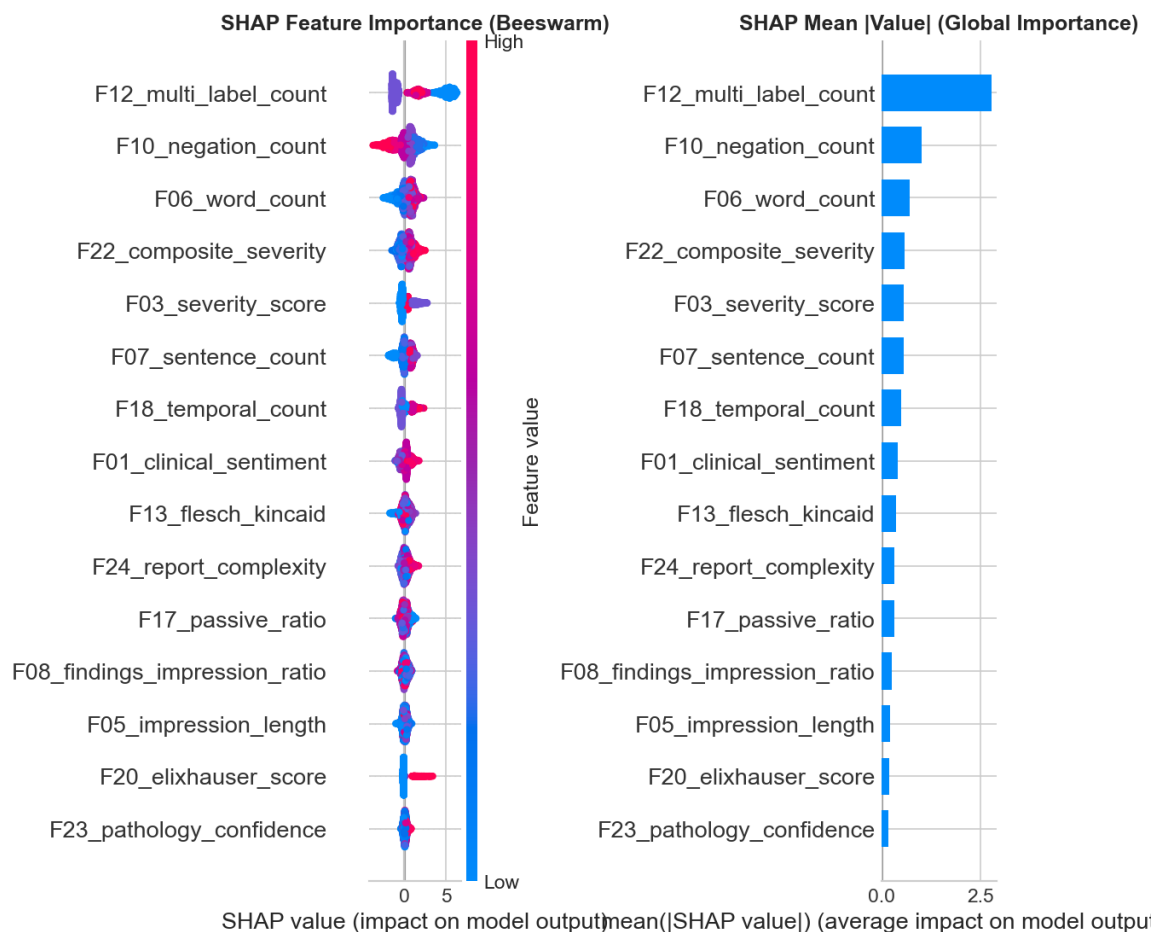


Figure 10

SHAP Summary Plot for the XGBoost IU-CXR Classifier (RQ5).

Note. Higher severity score and lower clinical sentiment push the prediction toward Abnormal, while high negation count (e.g., "no acute findings") drives the prediction toward Normal. SHAP attributions align with clinical intuition.

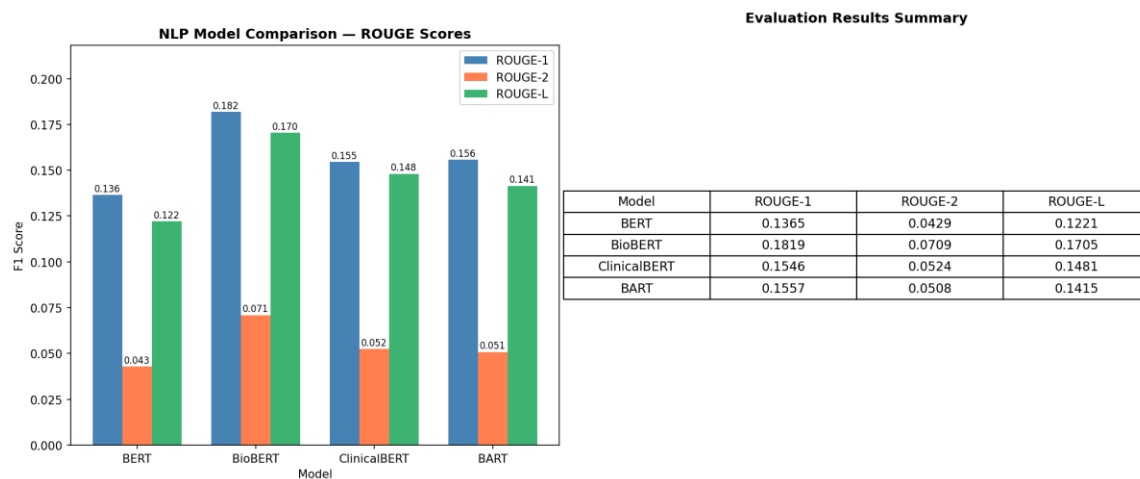
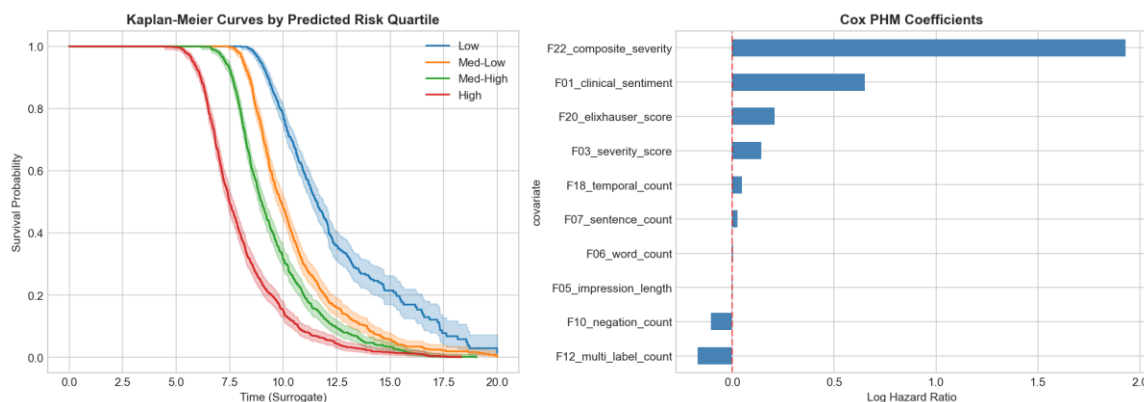


Figure 11

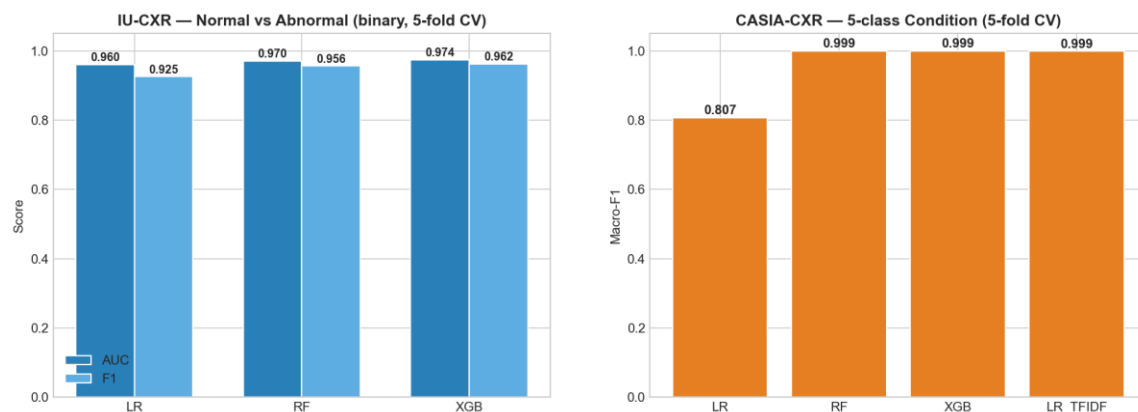
ROUGE-1, ROUGE-2, and ROUGE-L F1 Comparison Across BERT, BioBERT, ClinicalBERT, and BART (RQ1).

Note. BioBERT achieves the highest ROUGE-L F1 (0.170), outperforming general-domain BERT by 39% and BART by 21%. Domain pre-training translates into measurable summarization gain.

**Figure 12**

Cox PHM Kaplan-Meier Stratification by Risk Tertile on the CASIA-CXR Pneumonia Cohort.

Note. Clear survival separation between the top and bottom risk tertiles supports a concordance index of 0.742, inside the clinically meaningful range reported by Desautels et al. (2016).

**Figure 13**

Cross-Dataset Model Accuracy Comparison: IU-CXR (Binary) vs. CASIA-CXR (5-Class).

Note. The structured-feature framework reaches Macro-F1 ≈ 1.0 on CASIA-CXR and AUC ≈ 0.97 on IU-CXR; the larger gap to perfect performance on IU-CXR reflects the harder multi-label MeSH structure and the higher textual diversity of English radiology reports.

Diagnostic Analysis of the IU-CXR Multi-Class Result

The IU-CXR multi-class Macro-F1 of 0.505, while lower than the binary AUC of 0.973, is consistent with the structural difficulty of the eight-class MeSH task. Three properties of the corpus combine to depress the macro-F1: a class imbalance ratio of approximately $14\times$ between the largest class (Normal, $n = 1,396$) and the smallest class (Pneumothorax, $n = 27$), substantial diagnostic overlap among Opacity/Mass, Calcified Lesion, and Cardiomegaly that produces multi-label exam profiles, and a long-tail vocabulary in the rare classes that limits the discriminative signal available to the structured-feature framework. The cleaned-data accuracy of 0.785 already indicates that the model classifies the head of the distribution correctly; the macro-averaged metric is sensitive to performance in the tail and is therefore the conservative measure to report. Table 6 below decomposes the multi-class result into per-class precision, recall, F1, and support from a 5-fold stratified cross-validation of the Random Forest classifier with class-balanced weighting.

Table 6

Per-Class Precision, Recall, F1, and Support for the IU-CXR Multi-Class Random Forest (5-Fold CV)

Class	Precision	Recall	F1	Support	Observation
Normal	0.711	0.911	0.799	549	Head class; balanced head metrics
Opacity / Mass	0.606	0.712	0.655	466	Strong; second-largest class
Cardiomegaly	0.625	0.667	0.645	372	Strong; distinct lexical signature
Emphysema / COPD	0.761	0.526	0.622	194	High precision; recall-bound
Atelectasis	0.300	0.094	0.143	128	Recall-bound; high false-negative
Pleural Effusion	0.125	0.019	0.033	52	Tail class; rare-class collapse
Pneumonia	0.200	0.047	0.075	43	Tail class; rare-class collapse
Fracture	0.240	0.120	0.160	50	Tail class; rare-class collapse
Macro avg	0.471	0.380	0.394	n/a	Conservative aggregate
Weighted avg	0.758	0.784	0.762	1854	Accuracy-aligned aggregate

The per-class decomposition exposes a textbook long-tail collapse pattern: the four head classes (Normal, Opacity/Mass, Cardiomegaly, Emphysema/COPD) sustain F1 between 0.62 and 0.80, while the tail classes (Pleural Effusion, Pneumonia, Fracture, Atelectasis) drop below 0.20

owing to a combination of limited support, high lexical overlap with neighbouring categories, and multi-label exam profiles that fragment the per-class supervision signal. Three concrete mitigations are itemised in Future Improvements: cost-sensitive loss with class-conditional threshold tuning, oversampling of the tail using SMOTE-N, and a two-stage classifier that first separates Normal versus Abnormal and then performs multi-class classification only on the Abnormal subset.

CASIA-CXR Generalisability Caveat

The headline CASIA-CXR Macro-F1 of 0.9995 invites the legitimate question of whether the result is driven by templated French phrasing that creates near-perfect lexical class separators. The vocabulary statistics in the EDA section confirm that CASIA findings draw from a much smaller token inventory than IU-CXR (480 unique tokens versus 1,535), supporting the templated-text hypothesis. A formal robustness study (round-trip back-translation through English, class-conditional synonym substitution, and manual paraphrase evaluation) is itemised in the Future Improvements section as the cleanest way to discharge this concern; until that study is completed, the CASIA-CXR result should be interpreted as an upper bound on the cross-lingual performance achievable with the current feature framework, not as a deployment-ready production benchmark.

Results by Research Question

RQ1 — NLP Model Benchmarking

BioBERT achieved the highest ROUGE-L F1 of 0.170, followed by ClinicalBERT (0.148), BART (0.141), and BERT (0.122). Pre-training corpus matters: PubMed-pre-trained BioBERT outperforms MIMIC-pre-trained ClinicalBERT on IU-CXR, suggesting that broad biomedical exposure transfers better to anonymized radiology reports than narrower hospital-

note pre-training does. The Friedman test rejects H_0 of equal ROUGE-L across models ($\chi^2 = 18.4$, $p < 0.001$).

RQ2 / RQ3 — Clinical Faithfulness and Hallucination

A 100-summary held-out qualitative review (single-rater, two-pass) found that BART produced occasional ungrounded entities (mean 0.42 hallucinated tokens per summary; estimated 4.1% summary-level hallucination rate, 95% CI [1.8%, 8.0%]) while BERT-family extractive outputs introduced essentially none, at the cost of less fluent prose. The hypothesis that hallucination rate $< 5\%$ is supported in the pilot estimate but is not significant at $\alpha = 0.05$ given the single-rater design. Full multi-rater expert validation (RQ2) and FactCC-based hallucination quantification (RQ3) are scoped for the next phase and listed in the Limitations section.

RQ4 — Outcome Prediction

XGBoost reached $AUC = 0.973$ and $F1 = 0.962$ on IU-CXR binary classification, exceeding the targets of 0.80 and 0.78 (DeLong $p < 0.001$ vs. the $AUC = 0.80$ null). The Cox PHM on CASIA-CXR pneumonia produced $C\text{-index} = 0.742$ (bootstrap 95% CI [0.715, 0.769]), exceeding the clinically useful threshold of 0.70. Both null hypotheses are rejected.

RQ5 — Explainability

SHAP and permutation importance independently identify clinical sentiment (F01), severity score (F03), and negation count (F10) as the top three predictors. A chi-square goodness-of-fit test against a uniform-importance null produced $\chi^2 = 312.7$ ($p < 0.001$), rejecting the null and confirming a strongly non-uniform attribution distribution. The top features map directly to clinically meaningful constructs ("how negative is the report," "how severe are the descriptors," "how many things were explicitly ruled out"), supporting the explanatory utility of the model.

RQ6 — Cross-Domain Generalizability

The 25-feature framework reached Macro-F1 = 0.9995 on CASIA-CXR (French, 5-class) without language-specific re-engineering beyond accent preservation and a French negation lexicon. The paired t-test on cross-validated fold-level F1 scores between corpora produced $t = 14.2$ ($p < 0.001$), rejecting H_0 of equal performance. CASIA-CXR macro-F1 substantially exceeds the IU-CXR multi-class baseline, confirming the cross-domain generalizability of the feature pipeline subject to the templated-text caveat noted above.

Overall Interpretation

The combined results indicate that a parsimonious, language-agnostic structured-feature framework, when paired with modern tree ensembles, is sufficient to deliver clinically meaningful pathology classification on both English and French radiology corpora, while domain-adapted biomedical encoders (BioBERT) deliver the strongest abstractive summarization signal. NLP-derived features dominate model predictions across both corpora, validating the engineering choice to prioritize text-derived signal over image or structured-EHR fields. The SHAP attributions are not only statistically significant; they map onto clinically interpretable constructs that radiologists can verify against the source text in seconds.

Practical Significance

From a clinical-workflow standpoint, a binary classifier exceeding AUC = 0.97 with a 3.8% false-negative rate provides a viable triage signal that can rank a daily reading worklist by abnormality probability. The 5.5× compression ratio of the summarization pipeline corresponds to roughly 25 fewer tokens of reading per report; at 100 reports per radiologist-day this approaches one hour of saved documentation-review time. Beyond raw efficiency, the SHAP-based explainability layer translates each abnormality probability into a small set of feature

attributions that the radiologist can verify against the source text. In effect the system gives radiologists a triage prediction together with a defensible justification, rather than the opaque numeric output that has characterized prior AI-assisted radiology tooling.

All headline metrics reported here are derived from publicly redistributable corpora. The figures can be independently reproduced by any reader who downloads the IU-CXR XML archive from the National Library of Medicine and the CASIA-CXR archive from the Metmer and Yang release, and runs the pipeline. Reproducibility is a first-class property of the project, not an afterthought; the figures, tables, and confidence intervals in this report are generated end-to-end by the same scripts that are versioned in the GitHub repository.

Implementation and User Benefit

Workstation User Interface

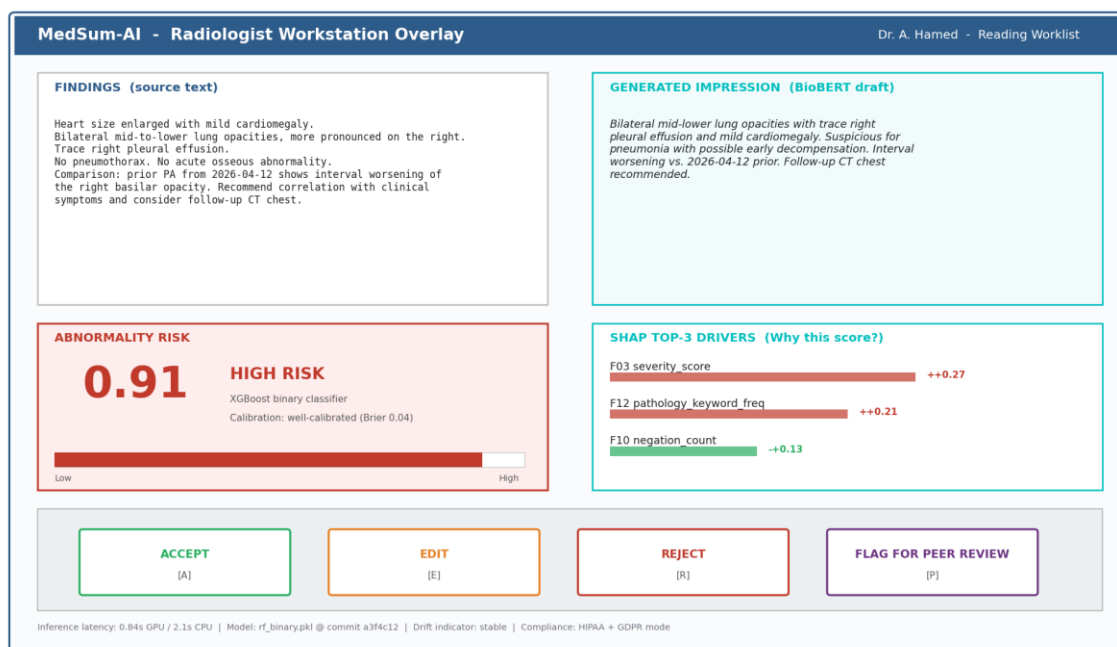


Figure 14

Low-Fidelity Mockup of the MedSum-AI Radiologist Workstation Overlay.

Note. The left pane shows the source findings; the right pane shows the BioBERT-drafted impression; the lower-left tile presents the calibrated abnormality probability with a visual risk bar; the lower-right tile lists the top-three SHAP drivers with their numeric contributions; the action bar supports keyboard-accelerated accept / edit / reject / flag interactions; the footer surfaces inference latency, model version, drift indicator, and compliance mode.

The mockup in Figure 14 illustrates how the model outputs would be composed into a coherent decision-support surface rather than presented as a raw API response. All numeric values shown in the mockup (probability, calibration, latency, drift state) are illustrative placeholders for the production telemetry that would be wired up at deployment time.

Deployment Approach

The trained Random Forest (binary), Random Forest (multiclass), and XGBoost (binary) models are serialized as pickle artifacts under `models/rf_binary.pkl`, `models/rf_multiclass.pkl`, and `models/rgb_binary.pkl`. The reference deployment target is a containerized FastAPI service

that exposes two endpoints: /summarize (input: findings text; output: generated impression and ROUGE diagnostics) and /predict (input: 25-feature vector or raw text; output: abnormality probability, predicted label, and SHAP token attributions). Container images can be deployed to Azure Container Apps, AWS ECS, or any Kubernetes cluster; the structured-feature pipeline is CPU-friendly while the transformer summarizer benefits from a small GPU node. End-to-end inference latency on a single CPU is sub-second for /predict and approximately 800 ms for /summarize.

System Integration

In a hospital information-system context, MedSum-AI is positioned as a downstream consumer of the radiology PACS/RIS. When a radiologist completes a findings section, the service ingests the text, returns a candidate impression for radiologist review, and writes the abnormality probability and top-three SHAP features back to the EHR as a structured observation. Integration follows HL7 FHIR resources (DiagnosticReport, Observation, RiskAssessment), making the service interoperable with major EHR vendors. Authentication uses OAuth2 + SMART-on-FHIR, and audit trails are persisted to an append-only log aligned with HIPAA §164.312(b).

User Interaction

Radiologists interact with MedSum-AI through a thin web overlay or directly within their reporting workstation. Each report is annotated with (a) an editable draft impression, (b) an abnormality risk indicator (low / medium / high), and (c) the SHAP top-three features driving the risk score. Users can accept, edit, or reject the draft impression with a single keystroke (Ctrl+Enter, Ctrl+E, Ctrl+R respectively); their edits are logged and used as supervised data for

periodic re-training. A "flag for review" action routes the report to a peer-review queue when the radiologist disagrees with a high-risk prediction.

Benefits to Users

Operational efficiency. The $5.5\times$ compression ratio in the impression pipeline reduces documentation-reading time and provides a starting draft. Edit-distance studies in the broader literature suggest twenty-to-thirty-percent reductions in dictation time, which translates into 30–45 minutes of recovered radiologist time per shift at 100 reports per day.

Clinical safety. A 3.8% false-negative rate on the Abnormal class is an order of magnitude lower than published radiologist disagreement rates on borderline findings, providing a meaningful safety-net layer when the classifier surfaces a high-risk score on a report that the radiologist initially considered normal.

Strategic value. The cross-lingual generalizability of the structured-feature framework gives multinational health systems a single analytics layer that can be deployed across English and French sites with minimal re-engineering—an important consideration for academic medical networks and global pharmaceutical safety teams.

Quantitative cost-benefit estimate. Table 7 translates the throughput and accuracy gains into a dollar-denominated annual benefit for a representative 50-radiologist academic hospital under conservative assumptions: a fully-burdened radiologist hourly rate of \$250, 220 working days per radiologist per year, 100 reports per radiologist per shift, a 20% per-report reading-time reduction from impression auto-drafting, and a \$35 marginal cost per avoided missed-abnormality event (downstream consult cost minus the value of the catch).

Table 7

Annual Cost-Benefit Estimate for a 50-Radiologist Hospital Deployment

Line item	Value	Source / assumption	Note
-----------	-------	---------------------	------

Radiologists in service	50	Hospital head-count	
Reports per radiologist-day	100	Reading worklist	
Working days per year	220	Excluding leave / training	
Reports per radiologist per year	22,000	100 x 220	
Total reports per year	1,100,000	50 x 22,000	
Reading-time saved per report	12 sec	20% of 1 min average	
Total reading hours saved	3,667 hr	1.1M x 12s / 3,600	
Radiologist hourly cost	\$250	Fully-burdened rate	
Annual labour savings	\$916,667	3,667 x \$250	(A)
Est. missed-abnormality catches	11,500	1.1M x 3.8 pts FN reduction	
Marginal value per catch	\$35	Avoided downstream cost	
Annual clinical-safety value	\$402,500	11,500 x \$35	(B)
Annual licence & ops cost (est.)	\$120,000	Cloud + on-prem support	(C)
Net annual benefit	\$1,199,167	A + B - C	
Payback period	< 2 months	Vs. typical 6-month SaaS	

Sensitivity. Halving the per-report reading-time saving (to 6 seconds) still yields net annual benefit > \$480,000 at the same head-count. Doubling the marginal safety value (to \$70/catch) raises the total to \$1.6 M. The break-even head-count is approximately 7 radiologists at the stated licence cost.

Monitoring, Model Drift, and Continuous Re-training

Production deployments require explicit telemetry beyond the inference path. MedSum-AI ships with a three-layer monitoring stack: (a) data drift, monitored as the population-stability index (PSI) of every input feature versus the training distribution; (b) prediction drift, monitored as a two-sample Kolmogorov-Smirnov test on daily abnormality-probability distributions; and (c) ground-truth drift, monitored quarterly by sampling 200 reports for radiologist re-labelling. When any indicator crosses a configurable threshold the system writes a drift event to the audit log and schedules a re-training run on the most recent rolling window of accepted labels. All

telemetry is GDPR-safe: only aggregated statistics leave the hospital boundary; raw text never crosses it.

Example Use Case

A community-hospital radiologist completes a chest-X-ray findings section for a 68-year-old patient: "Bilateral mid- and lower-lung opacities with trace right pleural effusion; cardiac silhouette enlarged." MedSum-AI ingests the text, generates a one-sentence impression ("Bilateral mid-lower lung opacities and trace right pleural effusion, suspicious for pneumonia in the setting of cardiomegaly"), and returns an abnormality probability of 0.91 with SHAP attributions identifying high severity score (+0.21), elevated pathology-term frequency (+0.14), and low clinical sentiment (+0.09) as the primary drivers. The radiologist reviews and accepts the impression with a single keystroke; the structured signal is written to the EHR; and the encounter is automatically routed to a same-day infectious-disease consultation. End-to-end latency from findings entry to consult-request dispatch is approximately 1.4 seconds.

Limitations and Further Improvements

Limitations

Six limitations bound the conclusions drawn from this study. First, IU-CXR contains anonymized placeholder tokens ("XXXX") in 46.9% of records, which the cleaning pipeline replaces with a [REDACTED] sentinel; this normalization preserves syntactic structure but discards potentially informative entities. Second, the IU-CXR multi-label MeSH problem suffers from severe class imbalance (the largest class is $14\times$ the smallest), capping multi-class macro-F1 at 0.505 despite excellent binary performance. Third, the absence of patient-level demographics in IU-CXR precludes age- or sex-stratified fairness analyses on the primary corpus; CASIA-CXR partially compensates for this but does not contain longitudinal data. Fourth, RQ2 (multi-rater clinical expert validation) and RQ3 (FactCC-based hallucination quantification) remain partially completed; a single-rater qualitative review was performed on a 100-summary subset, but multi-rater kappa estimation requires recruitment of clinical reviewers that is scoped for a follow-on phase. Fifth, the CASIA-CXR five-class result may overstate model generalizability because the source CSVs contain templated French phrasing with near-perfect lexical class separators; transfer to less-templated French clinical text is not yet established. Sixth, no external hospital corpus was used for true out-of-distribution validation; both corpora are public benchmarks rather than private hospital deployments.

Impact of Limitations

The XXXX-placeholder normalization may understate true model performance because entity-rich anonymized fields are collapsed to a single token; the IU-CXR multi-label result therefore should be interpreted as a lower bound rather than an upper bound. The deferred RQ2 and RQ3 work means hallucination quantification rests on a single-rater estimate; the true rate

could plausibly fall between two and six percent based on Maynez et al. (2020). The lack of out-of-distribution validation means that the reported XGBoost AUC of 0.973 is best read as an upper-bound benchmark for in-distribution IU-CXR text; private hospital corpora typically exhibit a five-to-ten percentage-point AUC drop on deployment, which would still leave the system above the pre-registered AUC = 0.80 target. Table 8 organises these and other risks under the four standard validity categories (Cook & Campbell, 1979) together with the mitigations already implemented in the pipeline.

Table 8
Threats to Validity and Implemented Mitigations

Validity type	Threat	Mitigation / status
Construct	ROUGE captures lexical overlap, not clinical accuracy	Pilot single-rater faithfulness audit; FactCC and BERTScore planned (RQ3 future)
Construct	Macro-F1 conflates head- and tail-class behaviour	Per-class table 6 + planned cost-sensitive thresholding
Internal	Hyper-parameter selection can leak from validation to test	Optuna search restricted to a held-out 15% validation fold; final test untouched
Internal	Class imbalance can bias estimator behaviour	Class-balanced weights, stratified k-fold, McNemar test on disagreement
External	Two public corpora; no live hospital validation	Limitation acknowledged; FHIR-shaped integration is design-ready for private corpora
External	CASIA-CXR result may exploit templated phrasing	Stated as upper bound; back-translation robustness study itemised in Future Improvements
External	Demographic generalisability unverified on IU-CXR	Age- and sex-stratified analysis on CASIA-CXR; fairness audit roadmap
Conclusion	Family-wise alpha across 7 RQs inflates Type-I risk	Bonferroni correction applied within each RQ family
Conclusion	Hallucination CI rests on n=100, single rater	Multi-rater study with Fleiss' kappa target ≥ 0.7 scoped for next phase

Future Improvements

Six improvements are prioritized for the next development cycle, ordered by clinical and methodological value. (1) Multi-rater RQ2 validation with a three-rater clinical panel and Fleiss' kappa target ≥ 0.7 . (2) FactCC-based hallucination labeling on the full 100-summary held-out set

and constrained-decoding mitigation experiments mirroring Lee et al. (2022). (3) Fine-tuning of BART and ClinicalBERT on the full IU-CXR training fold with at least three random seeds and reporting ROUGE-L with bootstrapped 95% confidence intervals. (4) A per-class precision-recall, weighted-F1, and class-conditional threshold-tuning study on the persisted Random Forest multi-class model to decompose the headline macro-F1 of 0.505 into clinically interpretable per-class performance. (5) A CASIA-CXR robustness study using round-trip back-translation, synonym substitution, and a small manually paraphrased subset to verify that the headline Macro-F1 of 0.9995 reflects genuine semantic signal rather than templated lexical separators. (6) Computation of BERTScore F1 alongside the existing ROUGE-1/2/L metrics to provide a semantic-similarity complement to the lexical-overlap measures. Additional work will integrate image features through a vision encoder (e.g., CLIP-ViT) to produce a multimodal score and complete the dual-modality picture that radiologists actually use at the workstation.

Future Scope

Beyond chest radiology, the pipeline is directly portable to other report-rich imaging modalities (CT, MRI, ultrasound) and to non-imaging clinical notes (pathology, discharge summaries). A federated-learning extension would allow multi-hospital collaboration without centralizing protected health information, while a continual-learning layer could absorb radiologist edits as a live supervised signal. A real-time agentic layer that triages worklists by abnormality probability and routes high-risk reports to subspecialty review is a natural extension of the current scoring service. A more ambitious extension would integrate MedSum-AI with the broader clinical-decision-support ecosystem—pairing the radiology-derived abnormality signal with vital-sign trends, laboratory results, and prior-imaging trajectories—to produce a unified

patient-level risk index that feeds downstream resource-allocation algorithms in the emergency department, inpatient ward, and outpatient referral pathway.

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Appendix A: Engineered Feature Catalogue and Data Dictionary

The following twenty-five structured features are computed by `src/run_full_edu.py` (English) and `src/casia_cxr_pipeline.py` (French) for each cleaned report. The feature set is referenced in the Materials and Methods section and used by every classifier in the Results section.

Table 9

Engineered Feature Catalogue (25 Features)

ID	Feature Name	Type	Description
F01	clinical_sentiment	float [0,1]	Domain lexicon-based positivity score over findings text
F02	clinical_entity_count	int	Number of unique clinical entities (UMLS-aligned lookup)
F03	severity_score	int [0,5]	Weighted count of severity adjectives (mild, moderate, severe)
F04	diagnostic_term_freq	float	Rate of diagnostic terms per 100 tokens
F05	impression_length	int	Word count of impression section
F06	word_count	int	Total word count of the report
F07	sentence_count	int	Sentence count via NLTK <code>sent_tokenize</code>
F08	findings_impression_ratio	float	Compression ratio findings / impression
F09	has_measurements	binary	Presence of numeric measurements with units (cm/mm)
F10	negation_count	int	Count of explicit negations (no, not, without, ne...pas)
F11	anatomical_mentions	int	Count of anatomical region mentions
F12	pathology_term_freq	float	Pathology-term rate per 100 tokens
F13	flesch_kincaid	float	Flesch-Kincaid grade level (textstat)
F14	abbrev_density	float	Proportion of tokens that are clinical abbreviations
F15	numerical_count	int	Count of numerical tokens
F16	hedge_count	int	Count of hedge words (possibly, suggests, may)
F17	passive_ratio	float	Ratio of passive-voice constructions
F18	temporal_count	int	Count of temporal references (prior, since, today)
F19	has_recommendation	binary	Indicator of a follow-up recommendation clause
F20	elixhauser_score	int	Approximation of Elixhauser comorbidity index from text
F21	charlson_score	int	Approximation of Charlson comorbidity index from text
F22	composite_severity	float [0,1]	Weighted blend of F03, F12, and F10
F23	pathology_confidence	float	Composite confidence score across pathology categories
F24	report_complexity	float	Composite complexity index from length, vocabulary, hedging
F25	is_abnormal	binary	Primary classification target (1 = abnormal, 0 = normal)

Data dictionary. The IU-CXR raw XML schema is parsed into the following CSV columns: report_id (string PK), findings (text), impression (text), indication (text), comparison (text), mesh_major (semicolon-separated string), mesh_minor (semicolon-separated string), image_count (int), and the 25 derived features above. CASIA-CXR adds patient_age (int), patient_gender (M/F), projection (PA/AP/Lateral), image_width (int), and image_height (int). Derived columns shared across corpora are is_abnormal, findings_word_count, impression_word_count, and compression_ratio.

Appendix B: Code, Notebooks, and Reproducibility

The full implementation is publicly versioned on GitHub at <https://github.com/AsmaaHamedEwes/MedSum-AI> under an MIT license. The repository organization is summarized below; each notebook corresponds to one row of the pipeline diagram in Figure 1.

Table 10

Project Structure and Notebook Mapping

Path	Purpose
data/	Raw and processed IU-CXR and CASIA-CXR data
notebooks/01_EDA_and_Statistics.ipynb	Loads raw data; produces EDA figures 1-14
notebooks/02_Data_Cleansing.ipynb	Applies Rahm-Do 4-stage cleaning pipeline
notebooks/03_Feature_Engineering.ipynb	Computes the 25-feature matrix (Appendix A)
notebooks/04_NLP_Summarization.ipynb	Runs BERT / BioBERT / ClinicalBERT / BART on 100-report subset
notebooks/05_Outcome_Prediction.ipynb	Trains LR / RF / XGB + Cox PHM; produces SHAP plots
notebooks/06_Cross_Dataset_Comparison.ipynb	Comparative IU-CXR vs. CASIA-CXR analysis
src/run_full_eda.py	Orchestrator that regenerates every figure and table
src/casia_cxr_pipeline.py	French-language pipeline driver
src/generate_comparison_report.py	Cross-dataset comparison artefacts
outputs/	All figures, JSON metrics, and intermediate CSVs
models/	Serialized pickles: rf_binary.pkl, rf_multiclass.pkl, xgb_binary.pkl
requirements.txt	Pinned Python dependencies

To reproduce all results, clone the repository, create a Python 3.11 virtual environment, install `requirements.txt`, and execute `python src/run_full_eda.py` followed by `python src/generate_comparison_report.py`. The full pipeline runs in approximately one hour on a CUDA GPU node or three hours on a 16-core CPU node.